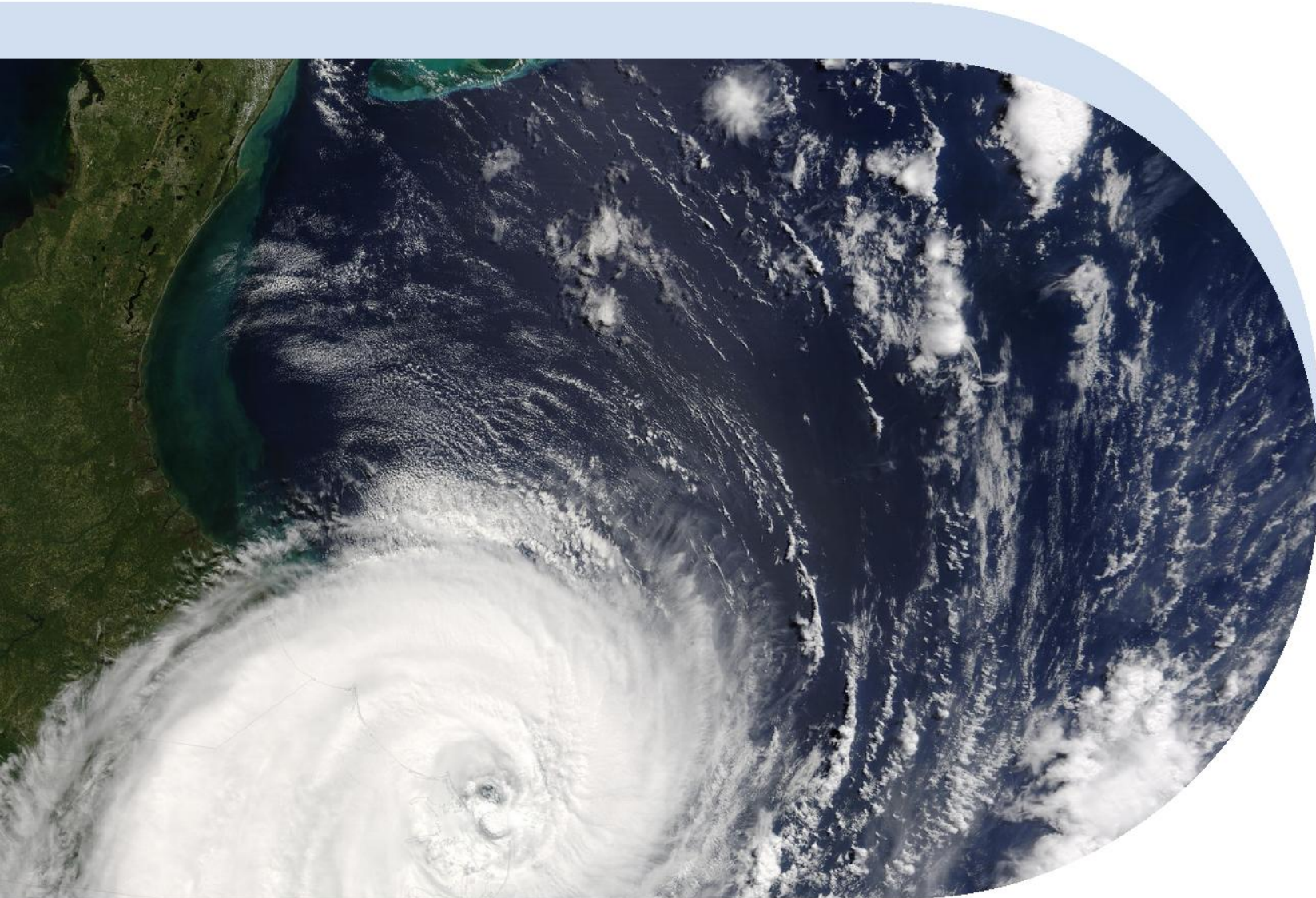


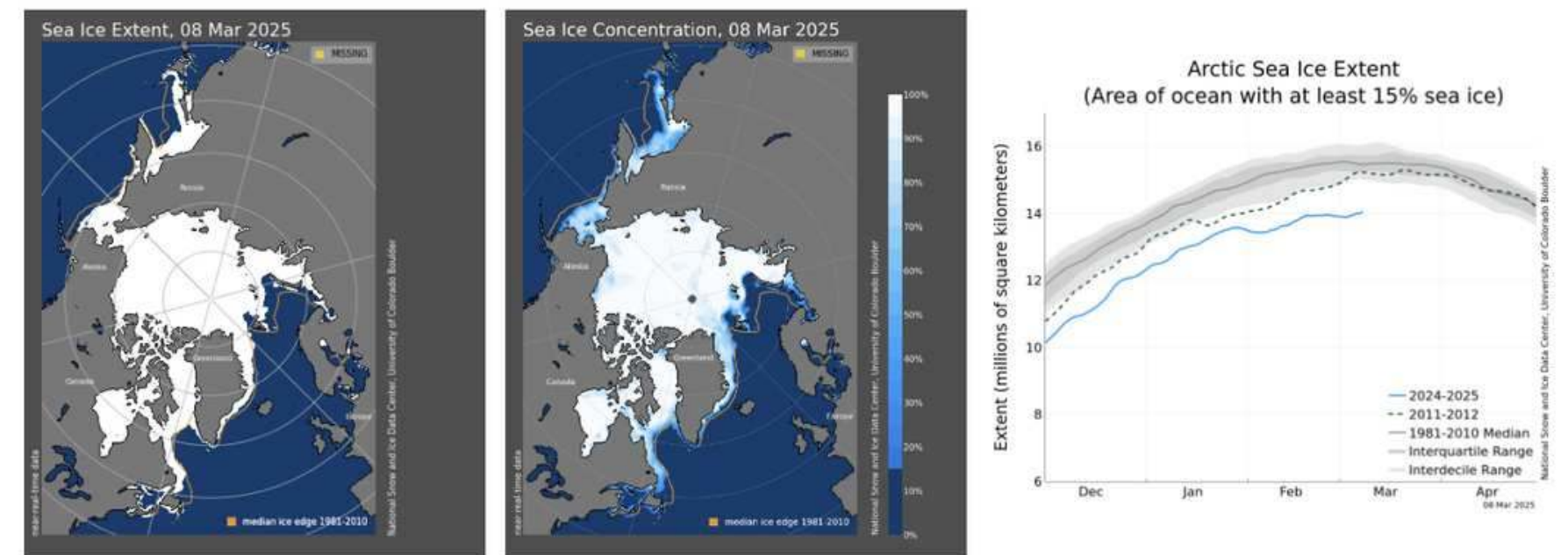


OPTIMAL ORBIT DESIGN FOR REMOTE SENSING OF CLIMATE CHANGE INDICATORS

Presented by MyoSat1

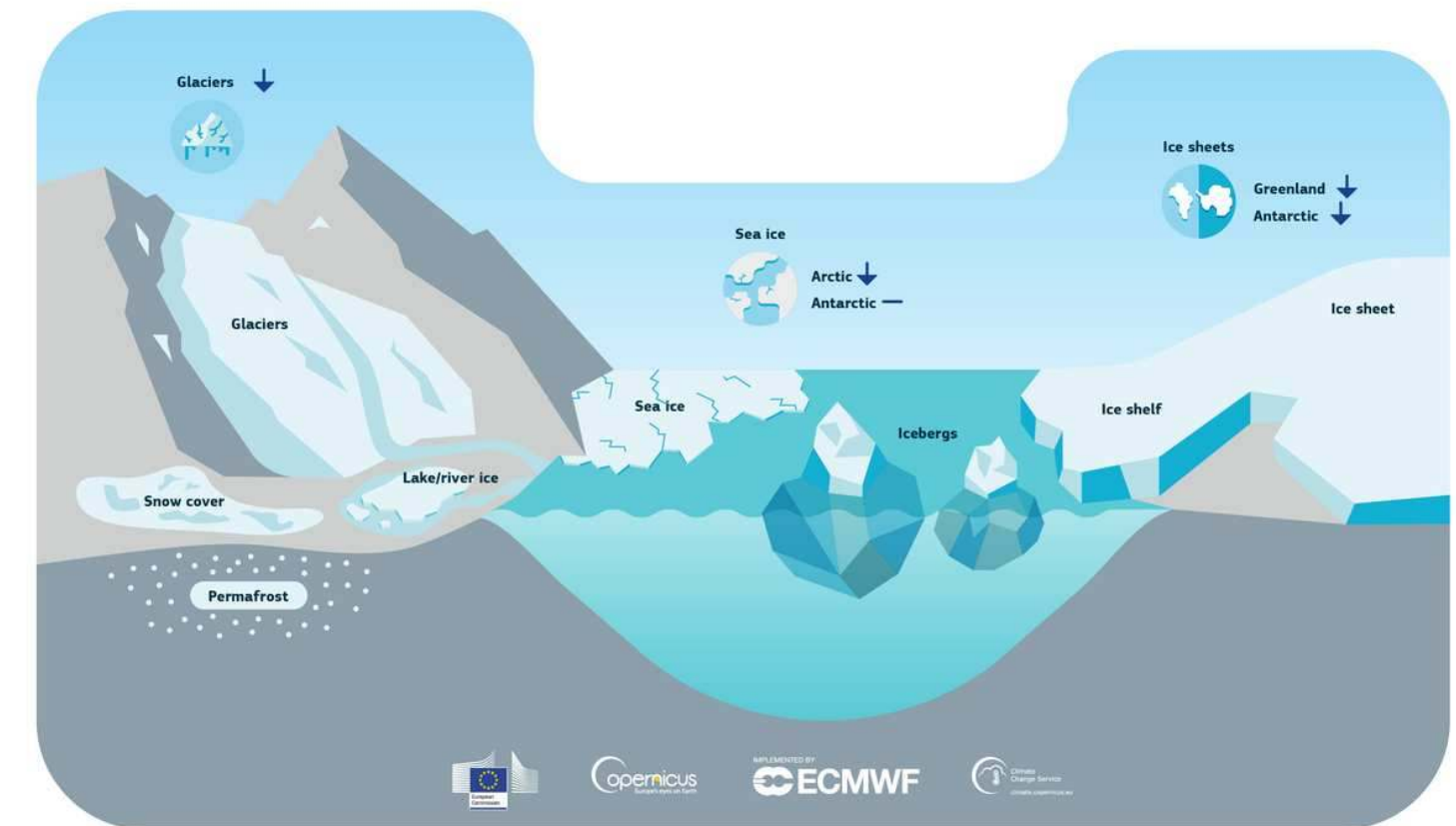
Introduction

The **Arctic ice** is melting rapidly due to global warming, making it one of the most important signs of climate change. Tracking these changes helps scientists understand **rising sea levels, temperature shifts, and weather patterns** that affect the whole planet. With improved satellite technology, we can monitor ice loss more effectively, predict future climate changes, and take action to protect our planet.



Why We Choose Crynosphere as our Climate Indicator?

- ♦ Sensitive to Climate Change – Ice sheets, glaciers, and sea ice respond rapidly to temperature shifts, making them key climate indicators.
- ♦ Long-Term Trends – Changes in ice extent, thickness, and volume provide clear evidence of global climate shifts over decades.
- ♦ Sea Level Impact – Melting glaciers and ice sheets contribute to rising sea levels, threatening coastal communities worldwide.
- ♦ Regulating Earth's Temperature – Ice plays a crucial role in the albedo effect, reflecting sunlight and helping regulate global temperatures.



Mission Focus Parameters

Coverage:

Target the polar regions (Arctic & Antarctic) where ice dynamics critically reflect climate change.

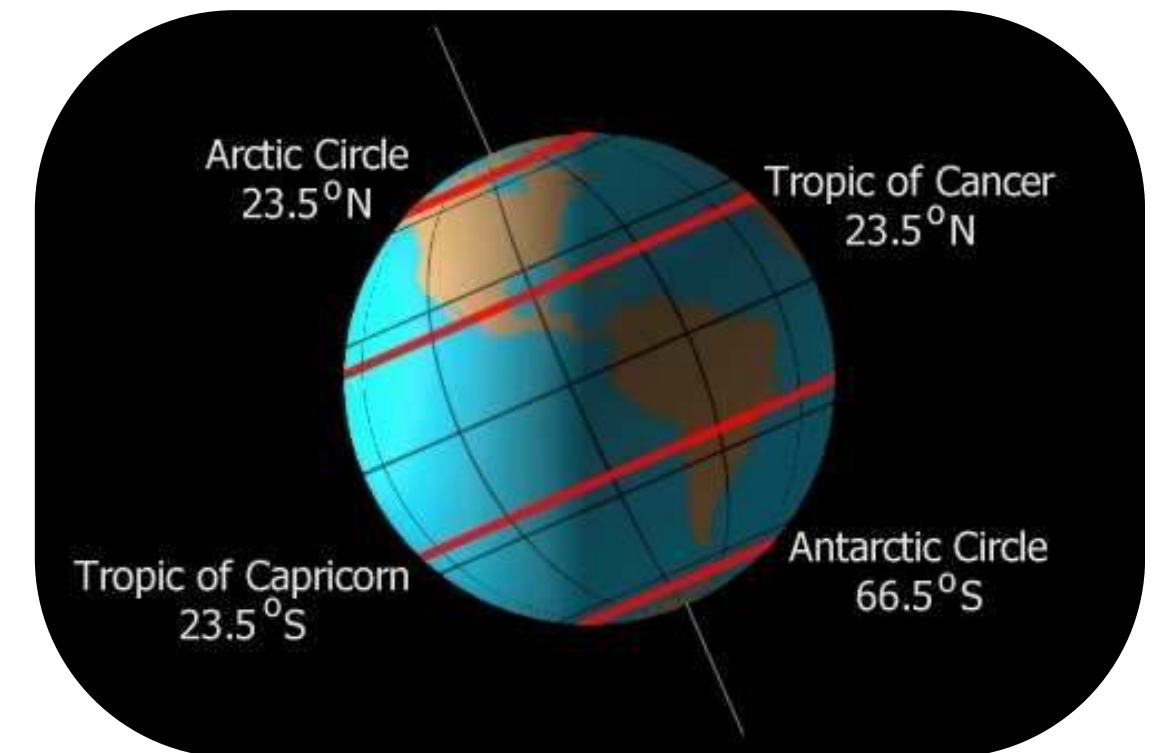
These regions encompass vast areas (e.g., Arctic ~20 million km², Antarctic ~14 million km²) with significant environmental impact.



Encompasses the Arctic Ocean
and surrounding lands across
North America, Europe, and Asia



Almost entirely covered by an ice sheet that holds approximately 90% of the world's fresh water.



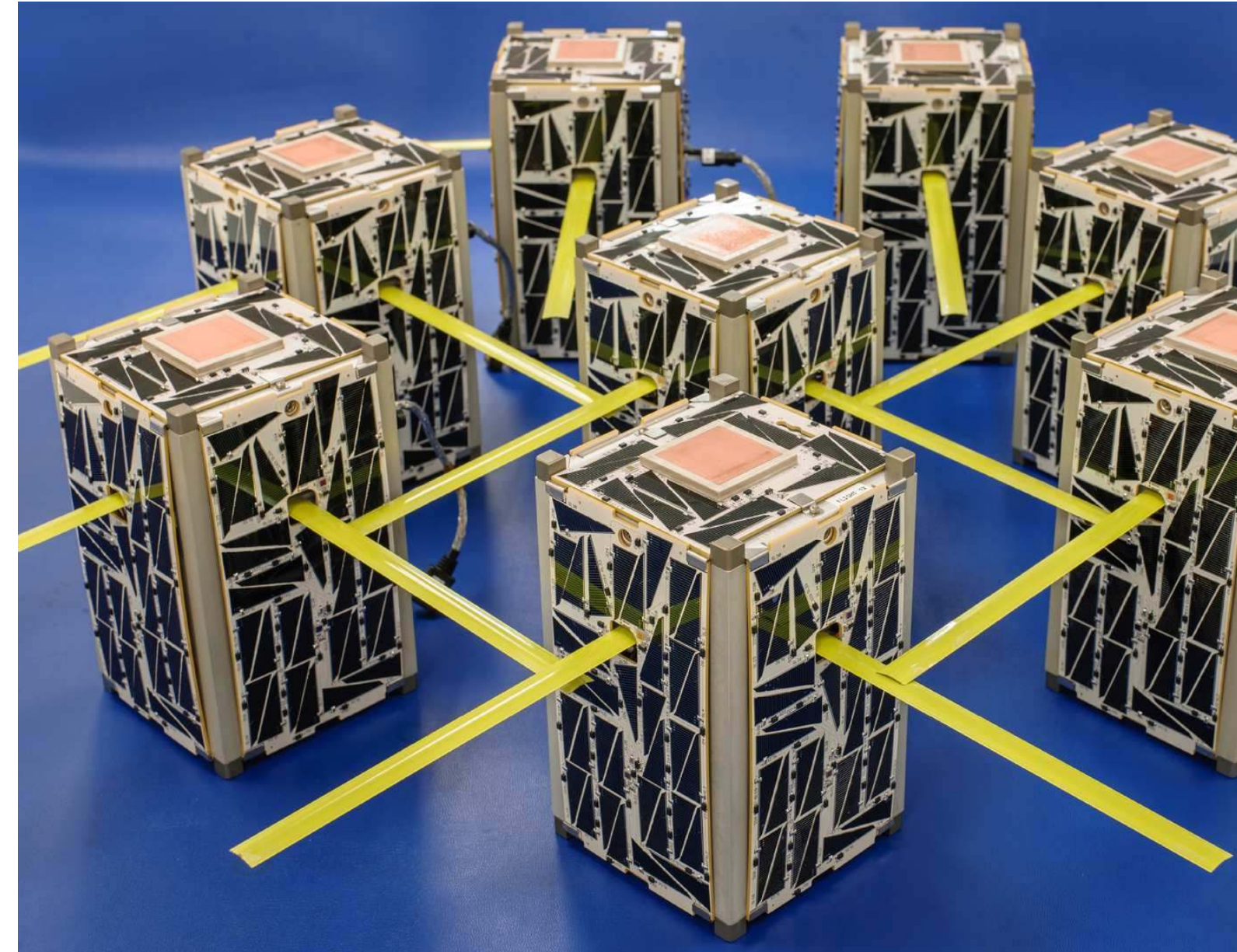
Mission Focus Parameters

- **Low Cost:** Emphasize a cost-efficient approach using small satellites. Affordable satellite platforms enable wider deployment and scalability.
- **Temporal Resolution:** High-frequency observations are crucial to capture rapid changes in ice extent and thickness. Reducing revisit times improves real-time monitoring and enhances the accuracy of climate models.



CubeSat Constellation Concept

- **CubeSat Constellation:** Deploy a network of 3U CubeSats optimized for polar coverage. Utilize a Walker Delta orbit configuration to ensure comprehensive and continuous monitoring over critical polar zones.



Mother CubeSat

CubeSats collect and relay climate data to a centralized Mother CubeSat

- **Onboard AI Processing:**

The Mother CubeSat aggregates, pre-processes, and prioritizes incoming data.

Reduces bandwidth requirements and optimizes transmission to ground stations.



MyoSat1 Orbit Details

Low Earth Orbit (LEO, 500-600 km)

Best for: Earth observation & IoT networks

Ice Observer CubeSats (LEO)

Reason for LEO:

- Closer to Earth for frequent observations of polar regions.
- High-resolution imaging of icebergs and glaciers.
- Low latency for quicker data transmission.



MyoSat1 Orbit Details

Sun-Synchronous Orbit (SSO, ~600 km)

Best for: Earth observation & imaging

Mother CubeSats (SSO)

Reason for SSO:

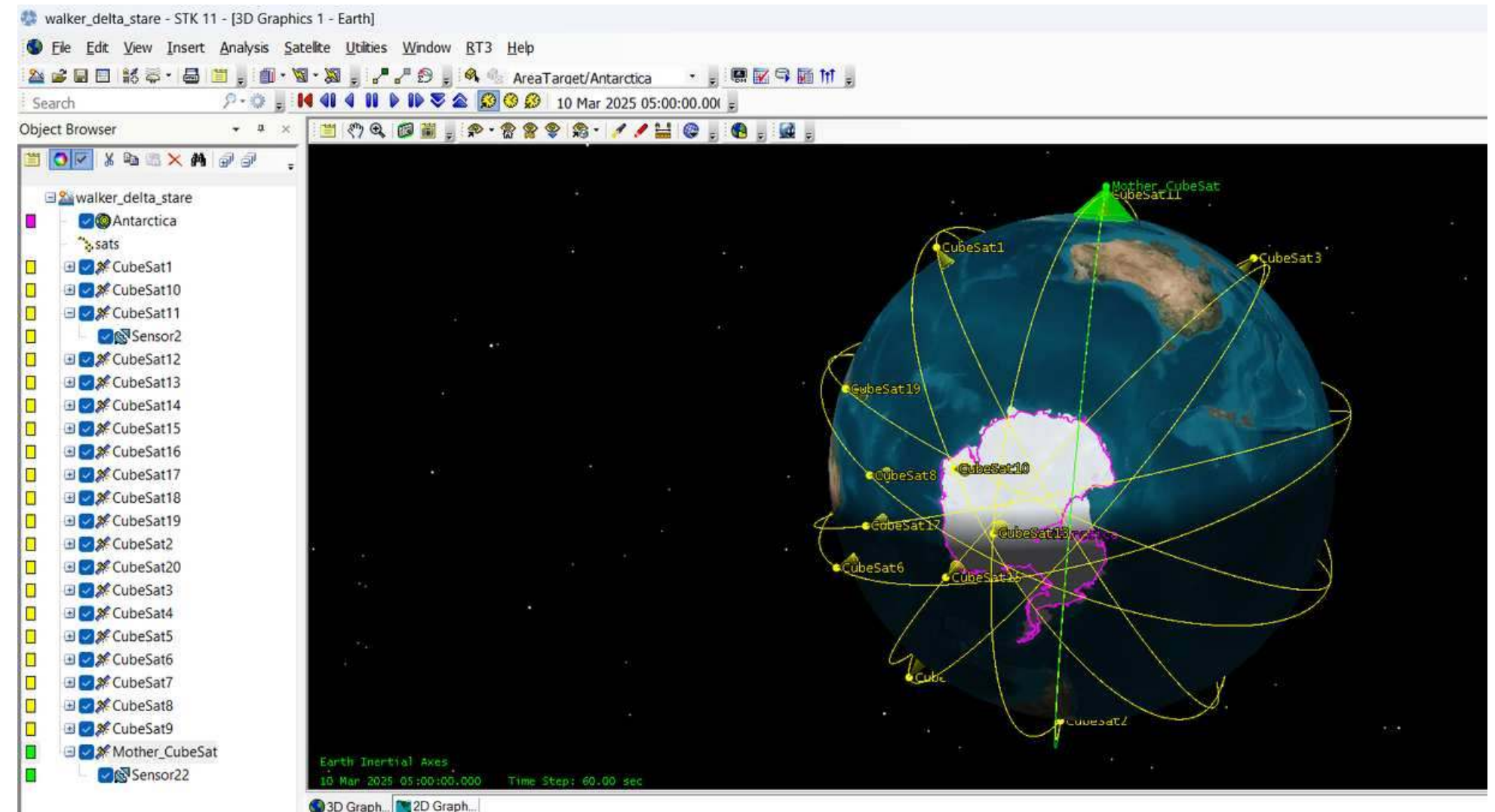
- Consistent sunlight for continuous power generation.
- Global coverage ensures efficient tracking of the Ice Observer CubeSats.
- Ideal for data relay, as SSO allows for regular communication with the CubeSats in LEO.



MyoSat1 Orbit Details

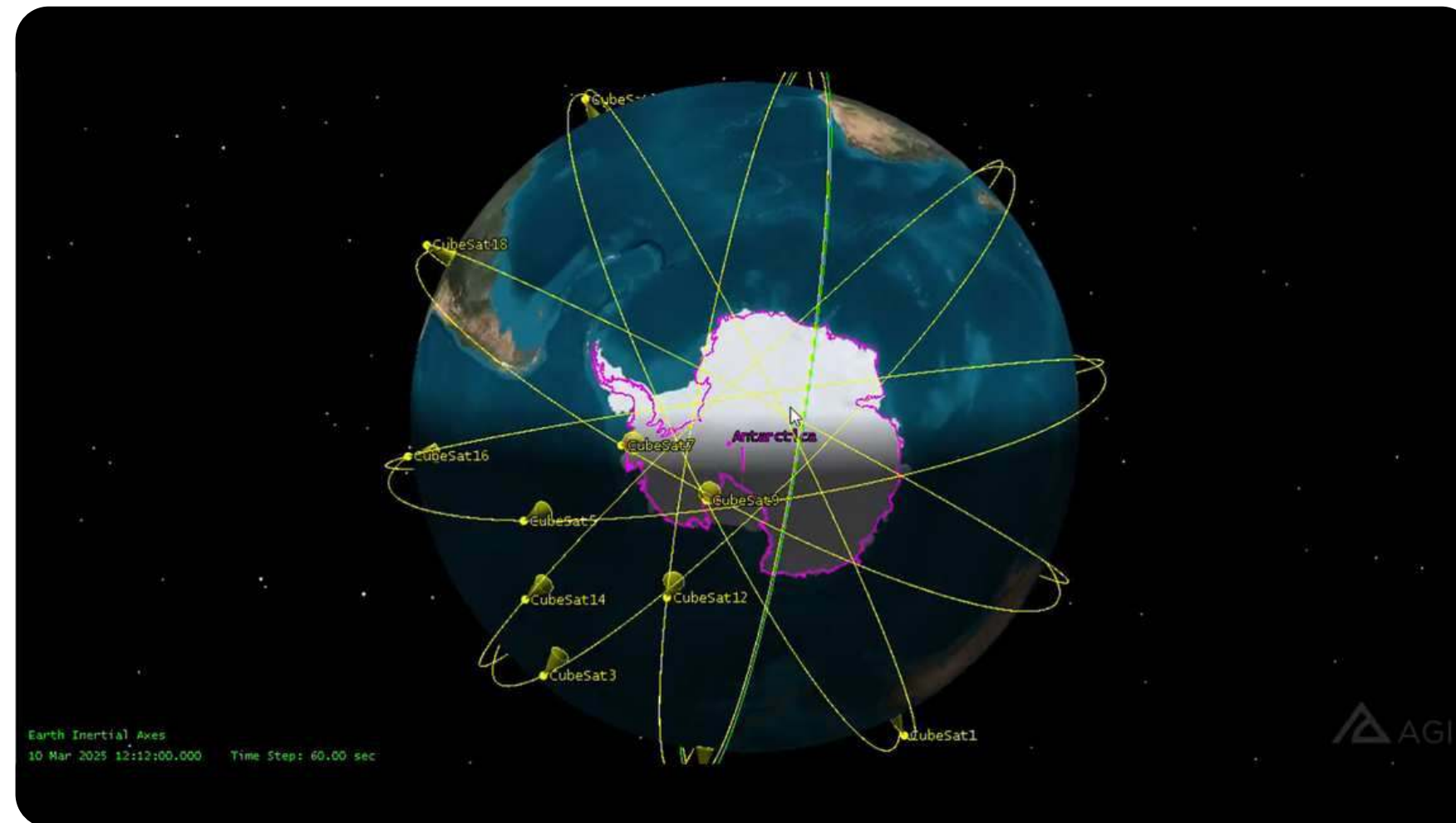
Walter-Delta Constellation

- Walker Delta (Walker- δ)
- Satellites are distributed evenly across multiple planes.
- Walker (20/10/1)
- Seed Satellite for Walker
 - LEO orbit
 - 500 km attitude
 - Inclination 97.8 degree
 - Eccentricity - 0 degree
 - RAAN - 0 degree
- Mother CubeSats
 - 700 km attitude (SSO), Same Inclination



Simulation

MyoSat1 Satellite Simulation



Youtube Link

[https://youtu.be/HsZAjDuPyqI?
si=x-wTBCt5AP8aE0FC](https://youtu.be/HsZAjDuPyqI?si=x-wTBCt5AP8aE0FC)

MyoSat1 Camera Choice

Gecko Imager

High-performance Gecko
Camera System for CubeSats



- Wide Coverage: The 160 km swath allows for wide-area coverage of both polar regions in a single pass.
- Compact and Efficient: The 1U camera fits within the limited payload space of a 3U CubeSat, ensuring high flexibility and miniaturization for missions with limited space.
- Optimized for Polar Orbits: The combination of the FOV and swath allows for efficient observation of polar regions, which is essential for monitoring phenomena such as icebergs, glaciers, and environmental changes.

39 M

Spatial Resolution

80 km

Swath at 500 km

<0.5 kg

Mass

1U

Physical Size

MyoSat1 Camera Choice

$$FOV = 2 \times \arctan\left(\frac{Swath}{2 \times h}\right)$$

- 20° FOV balances coverage area and image resolution.
- Provides a 160 km swath at 500 km altitude.
- Ideal for detailed monitoring of polar regions (e.g., icebergs and glaciers).
- Fits within a 1U camera on a 3U CubeSat, optimizing size, weight, and power (SWaP) constraints.
- Ensures efficient data collection and frequent, high-quality observations.

	RGB	WIDE SWATH
Spatial Resolution (GSD) at 500 km	39 m	39 m
Swath at 500 km	80 km	160 km
Mass (incl. electronics)	<0.5 kg	~ 0.5 kg
Spectral Bands	Standard RGB	Standard RGB
Physical Size	1U (65 mm x 90 mm x 105 mm)	1U (65 mm x 90 mm x 105 mm)
Data Format	8-bit or 10-bit	8-bit or 10-bit
Integrated Mass Data Storage	128 Gigabytes	128 Gigabytes
Compression	RAW or J2K lossless or lossy	RAW or J2K lossless or lossy
Data Interface	LVDS, SpaceWire	LVDS, SpaceWire
TMTC	SPI, I ² C, CAN, RS422, RS485	SPI, I ² C, CAN, RS422, RS485
Power Usage	Imaging mode <6 W Readout mode <4 W	Imaging mode <7 W Readout mode <4 W
Power Supply	5V DC	5V DC
Operating Temperature	+10°C to +30°C	+10°C to +30°C
Survival Temperature	-20°C to +55°C	-20°C to +55°C
TRL	9	9

Data Preprocessing

We intend to preprocess the data on mother CubeSat before sending to ground station

- Obtain RGB values from Gecko Camera.
- Train the image quality and cloud detection model to apply on mother cubesat.

Simulation Method

- Obtain the RGB values data and cloud score data from LandSat8 satellite(Surface Reflectance data).
- Train the image quality and cloud detection model



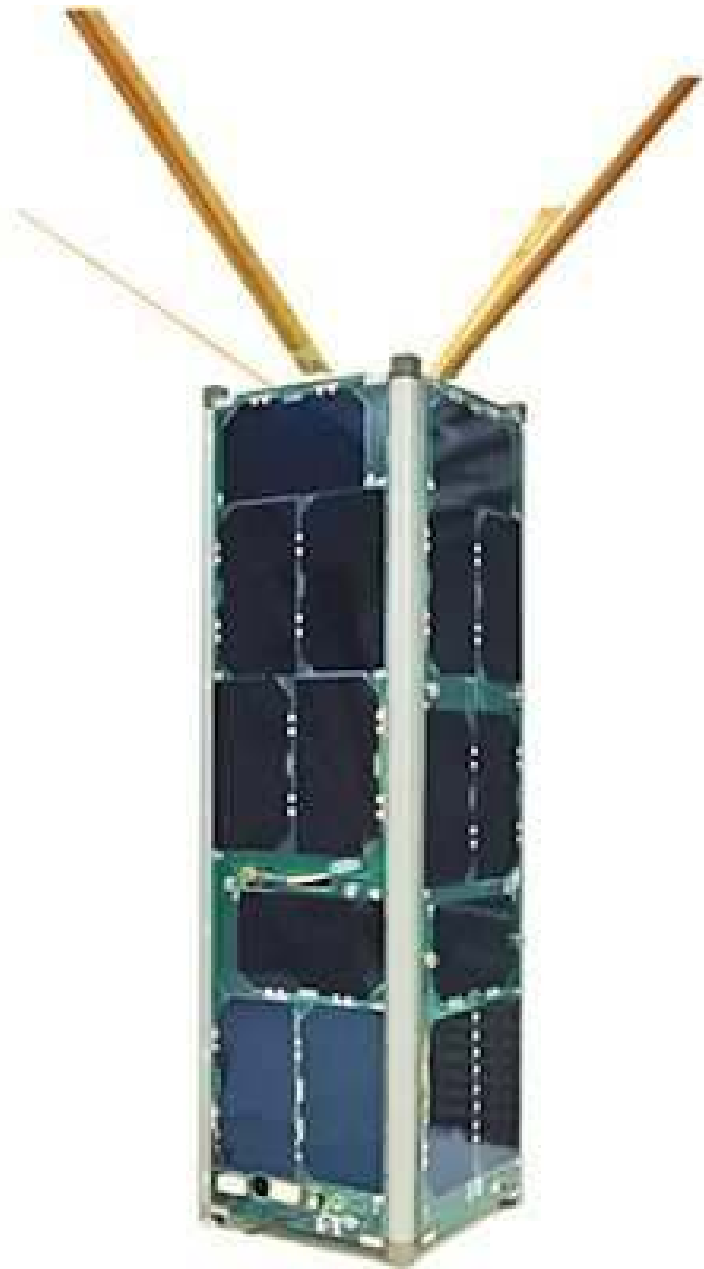
LandSat Image at Arctic Region(Surface Reflectance)

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▶ 18: "QA_RADSAT", unsigned int16, EPSG:32622, 8581x8631 px
▼ properties: Object (87 properties)
  ALGORITHM_SOURCE_SURFACE_REFLECTANCE: LaSRC_1.5.0
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  CLOUD_COVER: 94.02
  CLOUD_COVER_LAND: 93.67
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▼ Image LANDSAT/LC08/C02/T2_L2/LC08_010010_20200608 (19 bands)
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version: 1629906255754373
▼ bands: List (19 elements)
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  ▶ 1: "SR_B2", unsigned int16, EPSG:32622, 8581x8631 px
  ▶ 2: "SR_B3", unsigned int16, EPSG:32622, 8581x8631 px
  ▶ 3: "SR_B4", unsigned int16, EPSG:32622, 8581x8631 px
  ▶ 4: "SR_B5", unsigned int16, EPSG:32622, 8581x8631 px
```

LandSat Image at Arctic Region(Extracted image properties)

MyoSat1 Synthetic Aperture Radar (SAR)



X-band and C-band SAR instruments are feasible for use with 3U CubeSat platforms.

- active remote sensing systems that emit microwave signals to generate high-resolution images of Earth's surface, regardless of weather conditions or time of day.

Reference Link:

- https://www.politesi.polimi.it/retrieve/778b0c01-1821-4abe-8d2a-404fa7d5971b/2022_04_Kelleher_02.pdf?fbclid=IwZXh0bgNhZW0CMTEAAR0gKjXDpIKLOSfi7fAmhn8gF6XB7iJDUN7gBedt9Asv8eAhGPIQqMpV1dY_aem_GoKkyAU_kCA1IBMs6tUYdw
- <https://www.sciencedirect.com/science/article/pii/S0094576521003672>

X Band SAR

X-band SAR: Operates at frequencies between 8 to 12 GHz, corresponding to wavelengths of approximately 2.5 to 3.75 centimeters.

Resolution and Detail:

X-band SAR can achieve higher spatial resolutions, making it particularly suitable for applications requiring detailed imagery, such as urban mapping and infrastructure monitoring.

Penetration Capability:

The shorter wavelength results in limited penetration through vegetation and soil, making X-band less effective for applications needing subsurface information.

Notable Missions:

- TerraSAR-X and TanDEM-X: German satellites equipped with X-band SAR instruments, providing high-resolution imagery for various applications.
- Capella Space: Operates a constellation of small satellites with X-band SAR sensors, offering high-resolution, on-demand Earth observation data

C Band SAR

C-band SAR: Operates at frequencies between 4 to 8 GHz, corresponding to wavelengths of approximately 3.75 to 7.5 centimeters.

Resolution and Coverage:

Offers a balance between resolution and coverage area, making it versatile for various applications.

Penetration Capability:

With a longer wavelength than X-band, C-band SAR can penetrate through vegetation and certain soil types to some extent, making it useful for agricultural monitoring and soil moisture estimation.

Notable Missions:

- Sentinel-1: Part of the European Space Agency's Copernicus program, Sentinel-1 satellites carry C-band SAR instruments providing continuous all-weather, day-and-night imagery for land and ocean monitoring.
- RADARSAT-2: A Canadian satellite featuring a C-band SAR sensor, utilized for applications including disaster management, environmental monitoring, and resource mapping.

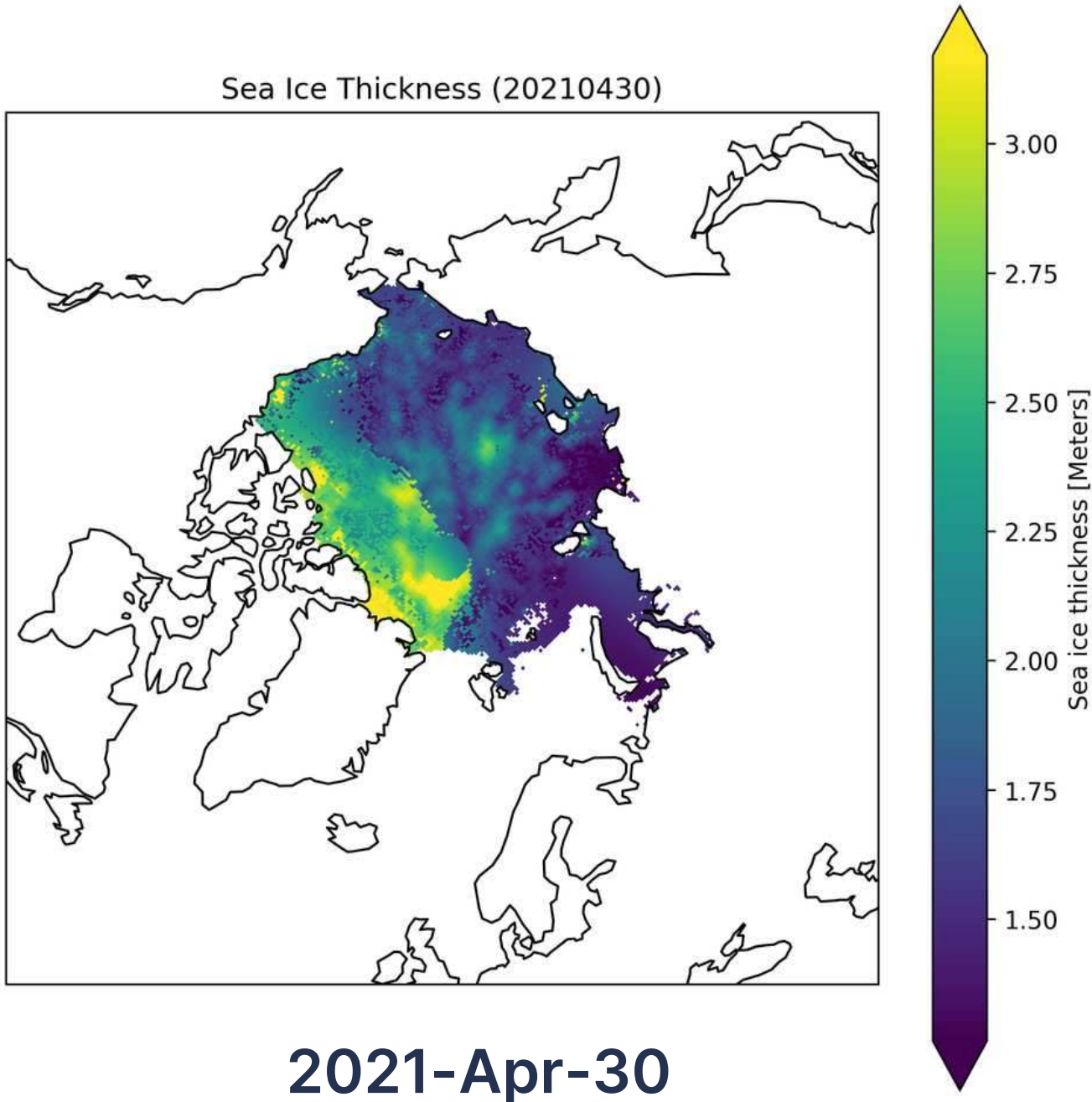
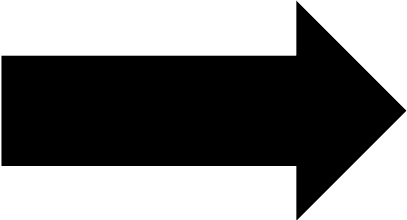
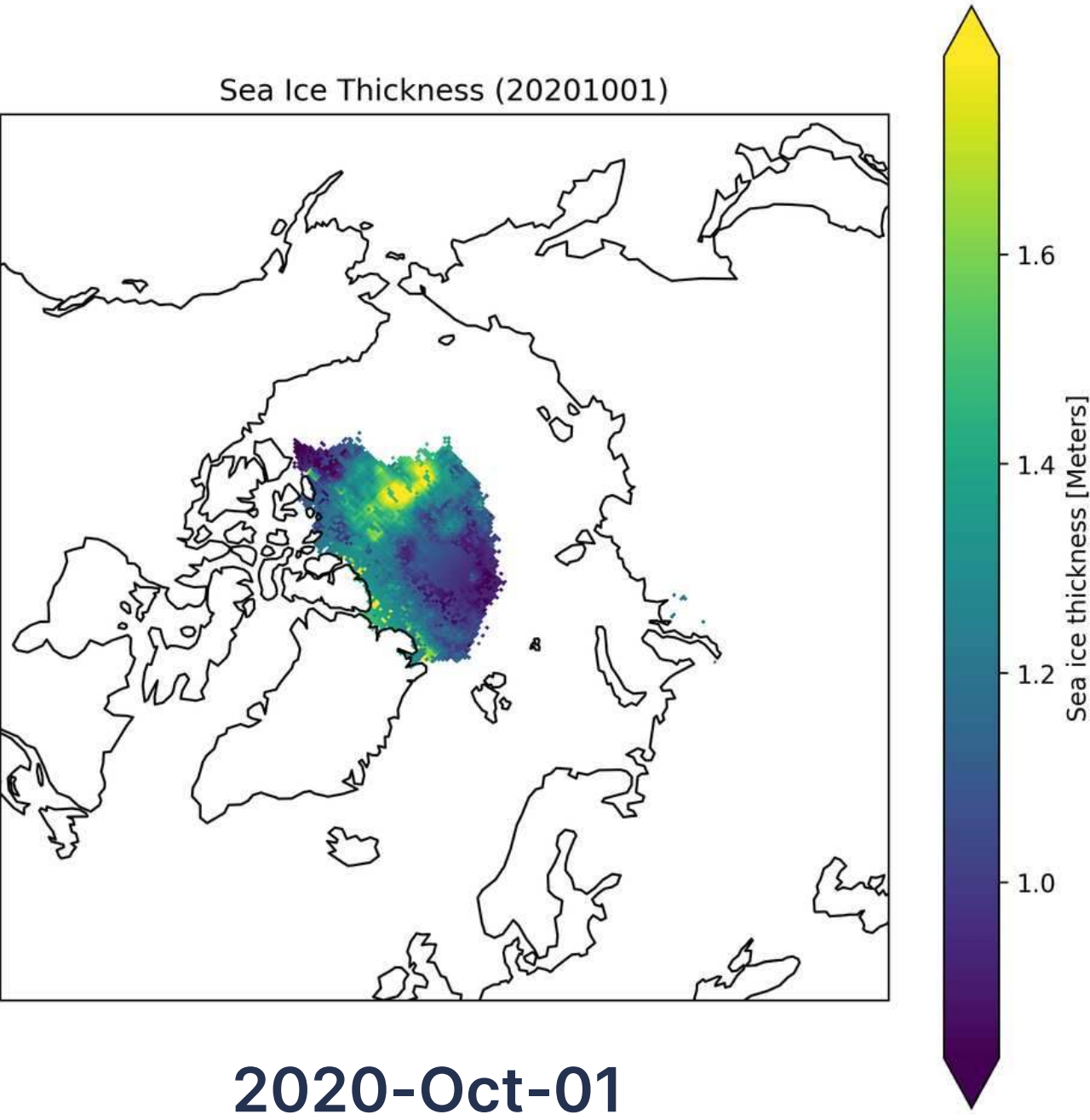
Sea Ice Thickness Data



CryoSat-2 Level-4 Sea Ice Elevation, Freeboard, and Thickness, Version 1 | National Snow and Ice Data Center

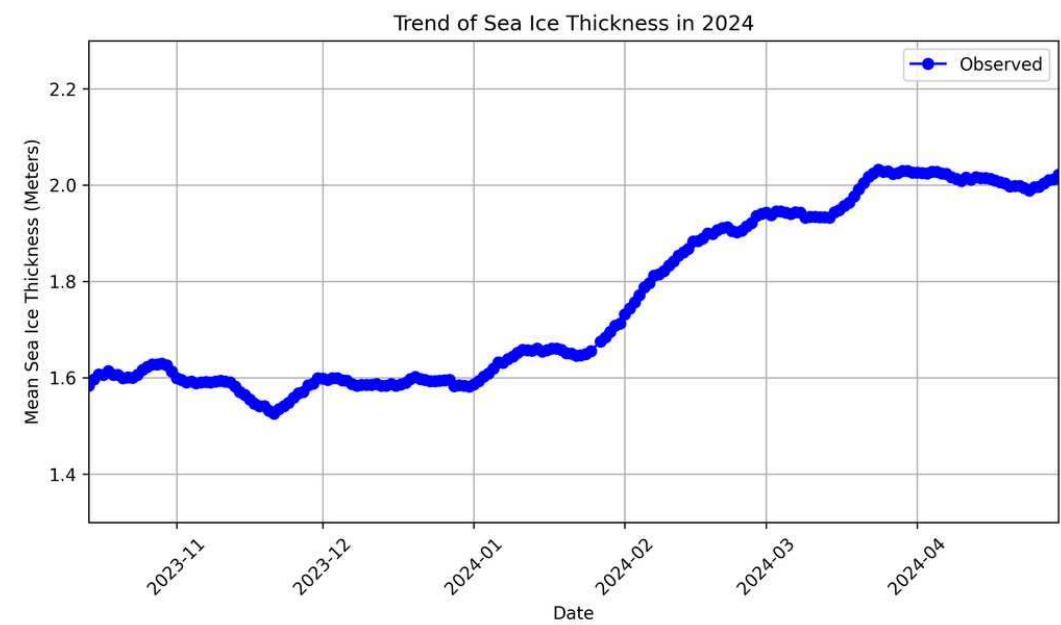
Data Citation and Acknowledgment As a condition of using these data, you must cite the use of this data set. Such a practice gives...

 nsidc.org

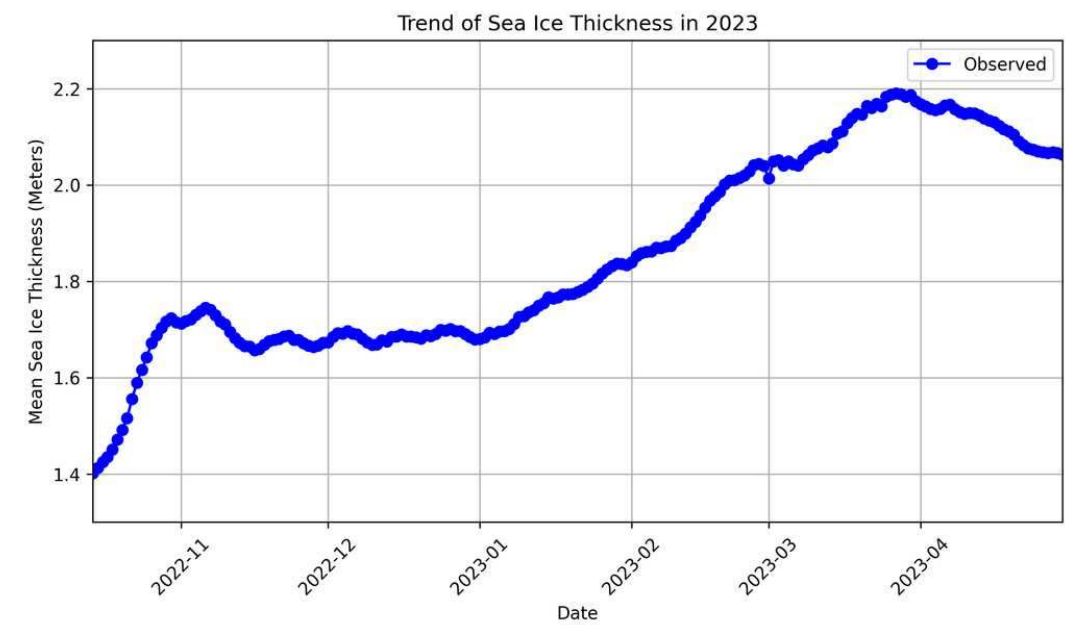


Sea Ice Data

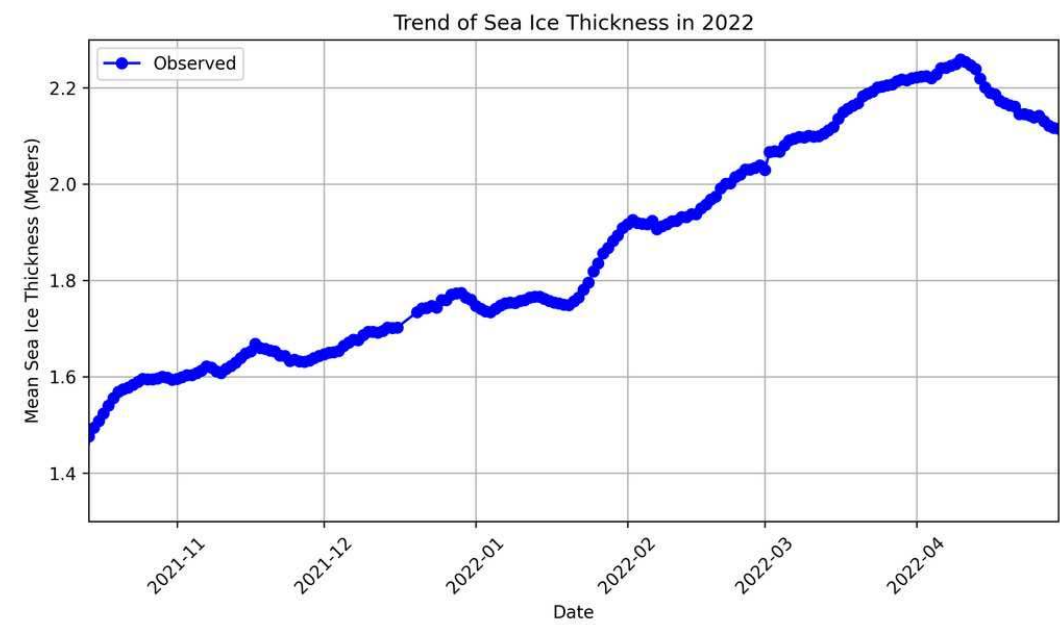
Trend



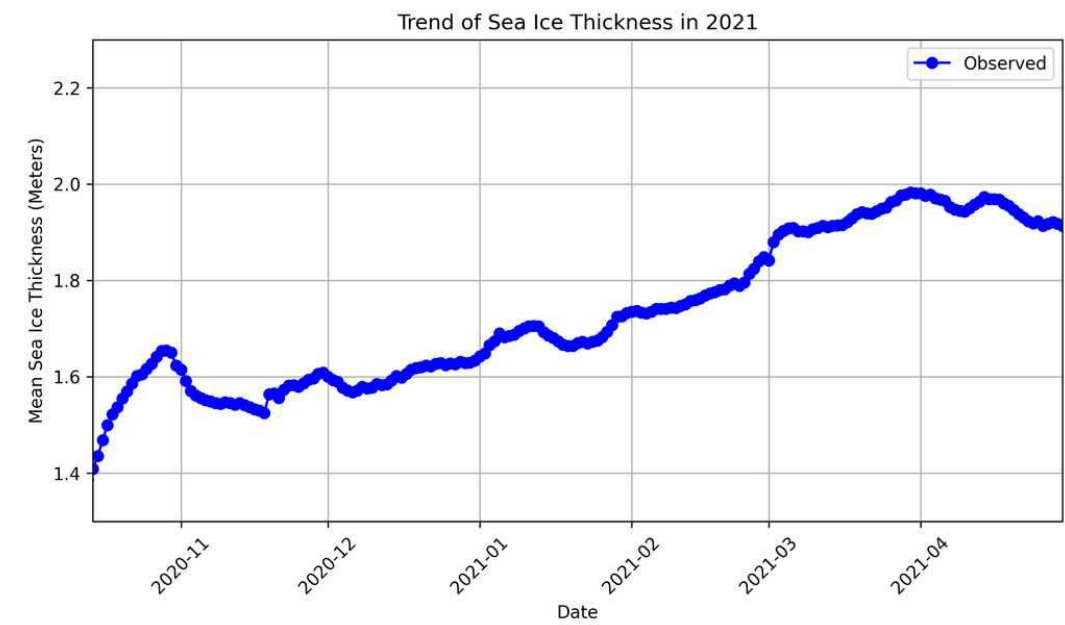
2024



2023

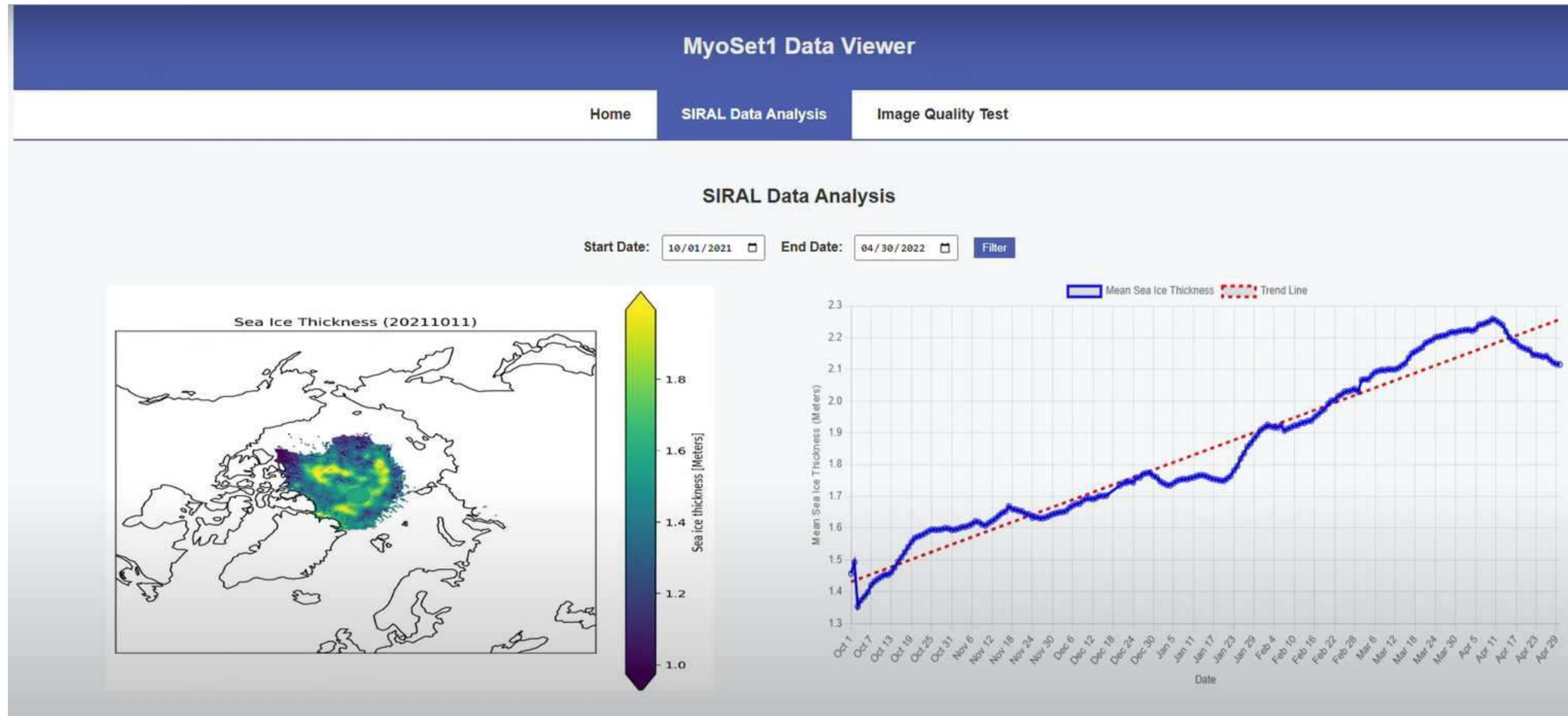


2022

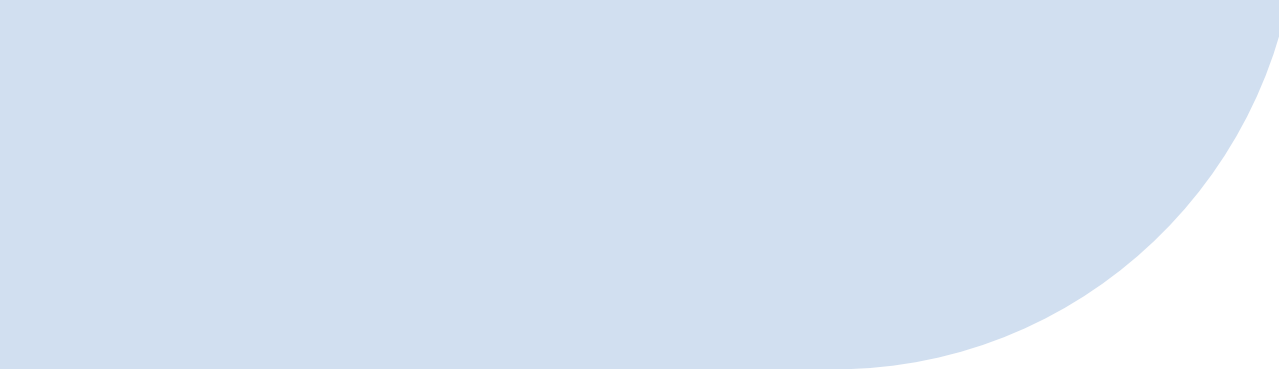


2021

Data Visualization & Trend Line Prediction



Youtube Link <https://youtu.be/GgFHHuLiqSw>



THANK YOU!

presented by MyoSat1

